

Theoretical Quantum Optics: Problem Sheet 6 (for 16.06.2025)

16. Decay of a macroscopic superposition state (1P)

Consider a harmonic oscillator that is described by the master equation

$$\dot{\rho} = \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger). \quad (1)$$

Assume that the harmonic oscillator is prepared in the pure state $\rho(t=0) = |\psi_0\rangle\langle\psi_0|$, where

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}(|n\rangle + |m\rangle) \quad (2)$$

is a superposition of two number states. Calculate the initial decay of the coherence $c_{n,m}(t) = \langle n|\rho(t)|m\rangle$, i.e., show that

$$\dot{c}_{n,m}(t) = -\frac{\Gamma_{n,m}}{2}c_{n,m}(t), \quad \text{for } t \rightarrow 0. \quad (3)$$

Derive the rate $\Gamma_{n,m}$.

17. Decay of a macroscopic superposition state: numerics (2P)

Write a Python program to simulate the master equation of a damped harmonic oscillator given in Eq. (1).

(a) Calculate the exact decay of $c_{n,m}(t)$ and compare it with the approximate expression

$$c_{n,m}(t) \simeq \frac{1}{2}e^{-\Gamma_{n,m}t/2}.$$

Choose $N_{\text{th}} \lesssim 5$ to keep the dimension of the Hilbertspace reasonably small.

(b) Evaluate $\rho(t)$ for times $\gamma t = [0.05, 0.5, 5]$ and $N_{\text{th}} = 3$ and visualize the decay of the initial superposition state into a thermal equilibrium state. Use, for example, the ‘pcolor’ function from matplotlib to show all the elements of the density operator in matrix form.

18. Quantum Regression Theorem (2P)

Consider a damped cavity mode, which is described by the master equation

$$\dot{\rho} = -i[H, \rho] + \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger) \quad (4)$$

with Hamiltonian

$$H = i\hbar\frac{g}{2}(a^2 - a^{\dagger 2}) + i\Omega(a - a^\dagger), \quad (5)$$

It is assumed that $g < \gamma/2$.

- (a) Derive the equations of motion for the expectation values of $\langle a(t) \rangle$ and $\langle a^\dagger(t) \rangle$. Based on this result, use the quantum regression theorem to derive a closed set of equations of motion for the correlation functions $\langle a^\dagger(\tau)a(0) \rangle$ and $\langle a(\tau)a(0) \rangle$.
- (b) Set $g = \Omega = 0$ and consider the correlation function $G^{(2)}(\tau) = \langle a^\dagger(0)a^\dagger(\tau)a(\tau)a(0) \rangle$. What is the value of $G^{(2)}(0) = 2N_{\text{th}}^2$ for a thermal equilibrium state? Use the quantum regression theorem to derive an expression for $G^{(2)}(\tau)$ for arbitrary τ .

19. Mollow triplet (1P)

By defining the vector of Pauli operators $\vec{S} = (\sigma_-, \sigma_+, \sigma_z)$, the optical Bloch equations for a driven two-level atom can be written as

$$\partial_t \langle \vec{S} \rangle = \mathbf{M} \langle \vec{S} \rangle - \vec{R}, \quad (6)$$

with a matrix $\mathbf{M}$ and a vector $\vec{R}$ derived in the lecture. With this definition the spectrum of incoherently scattered photons is given by

$$S_{\text{inc}}(\omega) = 2\Gamma \text{Re}(0, 1, 0) \left[\frac{1}{i(\omega - \omega_L)\mathbb{1} - \mathbf{M}} \right] \vec{C}(\tau = 0), \quad (7)$$

where $\vec{C}(0) = \langle \vec{S}\sigma_- \rangle_0 - \langle \vec{S} \rangle_0 \langle \sigma_- \rangle_0$ and $\langle \cdot \rangle_0$ denotes the steady-state expectation value.

- Write a Python program to evaluate and plot $S_{\text{inc}}(\omega)$ for arbitrary system parameters.
- Plot $S_{\text{inc}}(\omega)$ for a resonantly driven atom, $\Delta = 0$, and for increasing values of $\Omega/\Gamma = 0.1, 1, 2, 5, 10$.
- Plot and discuss $S_{\text{inc}}(\omega)$ for $\Omega/\Gamma \approx 5$ and different values of $\Delta/\Gamma = 0, 5, 10$. How do the positions and relative weights of the scattering peaks depend on the detuning?

Hint: For this exercise you can also use the code from the ‘Quantum Optics with Python’ notes as a starting point.

18. Quantum Regression Theorem (2P)

Consider a damped cavity mode, which is described by the master equation

$$\dot{\rho} = -i[H, \rho] + \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger) \quad (4)$$

with Hamiltonian

$$H = i\hbar \frac{g}{2}(a^2 - a^{\dagger 2}) + i\Omega(a - a^\dagger), \quad (5)$$

It is assumed that $g < \gamma/2$.

1

(a) Derive the equations of motion for the expectation values of $\langle a(t) \rangle$ and $\langle a^\dagger(t) \rangle$. Based on this result, use the quantum regression theorem to derive a closed set of equations of motion for the correlation functions $\langle a^\dagger(\tau)a(0) \rangle$ and $\langle a(\tau)a(0) \rangle$.

(b) Set $g = \Omega = 0$ and consider the correlation function $G^{(2)}(\tau) = \langle a^\dagger(0)a^\dagger(\tau)a(\tau)a(0) \rangle$. What is the value of $G^{(2)}(0) = 2N_{\text{th}}^2$ for a thermal equilibrium state? Use the quantum regression theorem to derive an expression for $G^{(2)}(\tau)$ for arbitrary τ .

$$\{a^\dagger, \hat{a}\}a = a^\dagger \{a, a\} \\ \text{tr}(\rho \{a^\dagger, \hat{a}\}a) = \text{tr}(\rho \{a^\dagger a - a a^\dagger + a^\dagger a - a^\dagger a\})$$

19. Mollow triplet (1P)

$$\langle \dot{a} \rangle(t) = \text{Tr}(a \dot{\rho}(t)) = \text{Tr}(a [-i[H, \rho] + \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger)])$$

$$\begin{aligned} &= -\frac{i}{\hbar} \text{Tr}(a [H, \rho]) + \frac{\gamma}{2}(N_{\text{th}} + 1) \text{Tr}(2\hat{a} \hat{a}^\dagger \hat{a} \rho - \hat{a} \hat{a}^\dagger \hat{a} \rho - \hat{a} \rho \hat{a}^\dagger \hat{a}) + \frac{\gamma}{2}N_{\text{th}} \text{Tr}(2\hat{a} \hat{a}^\dagger \hat{a} \rho - \hat{a} \hat{a}^\dagger \hat{a} \rho - \hat{a} \rho \hat{a}^\dagger \hat{a}) \\ &= \text{Tr}(a H \rho - a \rho H) \\ &= \text{Tr}(a H \rho - H a \rho) \\ &= -\langle [H, a] \rangle \end{aligned}$$

$$\begin{aligned} &= \text{tr}(2\hat{a}^\dagger \hat{a} \hat{a} \rho - a a^\dagger a \rho - a^\dagger a a \rho) \\ &= \text{tr}(\hat{a}^\dagger \hat{a} \hat{a} \rho - \hat{a} \hat{a}^\dagger \hat{a} \rho + \hat{a}^\dagger \hat{a} \hat{a} \rho - \hat{a}^\dagger \hat{a} \hat{a} \rho) \\ &= \langle [\alpha^\dagger, \alpha] \rangle = -1 \end{aligned}$$

$$= \frac{i}{\hbar} \langle [H, a] \rangle + \frac{\gamma}{2}(N_{\text{th}} + 1) \langle [\hat{a}^\dagger, \hat{a}] \hat{a} + \hat{a}^\dagger [\hat{a}, \hat{a}] \rangle + \frac{\gamma}{2}N_{\text{th}} \langle [a, a^\dagger] \hat{a}^\dagger + \hat{a} [a, a^\dagger] \rangle$$

$$= \frac{i}{\hbar} \langle [H, a] \rangle + \frac{\gamma}{2}(N_{\text{th}} + 1) \langle -\hat{a} + 0 \rangle + \frac{\gamma}{2}N_{\text{th}} \langle 0 + \hat{a} \rangle$$

$$= \frac{i}{\hbar} \langle [H, a] \rangle - \frac{\gamma}{2}(N_{\text{th}} + 1) \langle \hat{a} \rangle + \frac{\gamma}{2}N_{\text{th}} \langle \hat{a} \rangle$$

$$= \frac{i}{\hbar} \langle [H, a] \rangle - \frac{\gamma}{2} \langle \hat{a} \rangle = -\frac{1}{\hbar} (\hbar g \langle \hat{a}^\dagger \rangle + \hbar \Omega) - \frac{\gamma}{2} \langle \hat{a} \rangle = -g \langle \hat{a}^\dagger \rangle - \frac{\gamma}{2} \langle \hat{a} \rangle - \Omega$$

$$H = i\hbar \frac{g}{2}(a^2 - a^{\dagger 2}) + i\Omega(a - a^\dagger),$$

$$\begin{aligned} [H, a] &= i\hbar \frac{g}{2}([a^2, a] - [\hat{a}^{\dagger 2}, a]) + i\Omega([a, a] - [\hat{a}^\dagger, a]) \\ &= i\hbar \frac{g}{2}(2a) + i\Omega \\ &= i(\hbar g \hat{a}^\dagger + \Omega) \end{aligned}$$

$$\begin{aligned} [H, \hat{a}] &= i\hbar \frac{g}{2}([a^2, \hat{a}] - [\hat{a}^{\dagger 2}, \hat{a}]) + i\Omega([a, \hat{a}] - [\hat{a}^\dagger, \hat{a}]) \\ &= i\hbar \frac{g}{2}(2\hat{a}) + i\Omega \\ &= i(\hbar g \hat{a} + \Omega) \end{aligned}$$

$$\text{Analogously: } \langle \dot{a}^\dagger \rangle = \frac{i}{\hbar} \langle [H, a^\dagger] \rangle + \frac{\gamma}{2}(N_{\text{th}} + 1) \langle [\alpha^\dagger, \hat{a}^\dagger] a + \alpha^\dagger [\hat{a}, a] \rangle + \frac{\gamma}{2}N_{\text{th}} \langle [a, \hat{a}^\dagger] \hat{a}^\dagger + a [\hat{a}^\dagger, a] \rangle$$

$$= -g \langle \hat{a} \rangle - \frac{\gamma}{2}(N_{\text{th}} + 1) \langle \hat{a} \rangle + \frac{\gamma}{2}N_{\text{th}} \langle \hat{a}^\dagger \rangle - \Omega$$

$$= -g \langle \hat{a} \rangle - \frac{\gamma}{2} \langle \hat{a}^\dagger \rangle - \Omega$$

$$\Rightarrow \langle \dot{O} \rangle(t) = \frac{i}{\hbar} \langle [H, O] \rangle + \sum_k \frac{\Gamma_k}{2} \langle [c_k^\dagger, O] c_k + c_k^\dagger [O, c_k] \rangle$$

$$\rightarrow \langle \dot{a} \rangle = -g \langle \hat{a}^\dagger \rangle - \frac{\gamma}{2} \langle \hat{a} \rangle - \Omega$$

$$\begin{pmatrix} \langle \dot{a} \rangle \\ \langle \dot{a}^\dagger \rangle \end{pmatrix} = \begin{pmatrix} -g & -\frac{\gamma}{2} \\ \frac{\gamma}{2} & -g \end{pmatrix} \begin{pmatrix} \langle \hat{a} \rangle \\ \langle \hat{a}^\dagger \rangle \end{pmatrix} - \Omega$$

$$\hat{A} = \begin{pmatrix} \langle \dot{a} \rangle \\ \langle \dot{a}^\dagger \rangle \end{pmatrix} \rightarrow \hat{\hat{A}} = \begin{pmatrix} -g & -\frac{\gamma}{2} \\ \frac{\gamma}{2} & -g \end{pmatrix} - \Omega$$

$$\langle a(t) \rangle = \frac{1}{2} \left[\left(\alpha + \beta + \frac{2\Omega}{g + \frac{\gamma}{2}} \right) e^{-(g + \frac{\gamma}{2})t} + (\alpha - \beta) e^{(g - \frac{\gamma}{2})t} - \frac{2\Omega}{g + \frac{\gamma}{2}} \right]$$

$$\langle a^\dagger(t) \rangle = \frac{1}{2} \left[\left(\alpha + \beta + \frac{2\Omega}{g + \frac{\gamma}{2}} \right) e^{-(g + \frac{\gamma}{2})t} - (\alpha - \beta) e^{(g - \frac{\gamma}{2})t} - \frac{2\Omega}{g + \frac{\gamma}{2}} \right]$$

$$\frac{d}{dt} \langle O_i \rangle = \sum_j M_{ij} \langle O_j \rangle - \vec{R}_i$$

$$\rightarrow M_{ij} = - \begin{pmatrix} \gamma_{12} & g \\ g & \gamma_{12} \end{pmatrix}$$

$$\vec{R} = \Omega \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Then (for $t_1 > t_2$):

$$\frac{d}{dt_1} \langle O_i(t_1) O_k(t_2) \rangle = \sum_j M_{ij} \langle O_j(t_1) O_k(t_2) \rangle - \vec{R}_i \langle O_k(t_2) \rangle$$

use quantum regression theorem to derive
 $\langle a^\dagger(\tau) a(0) \rangle$ and $\langle a(\tau) a(0) \rangle$.

$$\frac{d}{d\tau} \langle a^\dagger(\tau) a(0) \rangle = \sum_j M_{2j} \langle O_j(\tau) a(0) \rangle - \Omega \langle a(0) \rangle$$

$$\begin{aligned} \hat{a}^\dagger &= O_2 \rightarrow i=2 \\ \hat{a} &= O_1 \rightarrow k=1 \end{aligned} = M_{21} \langle O_1(\tau) a(0) \rangle + M_{22} \langle O_2(\tau) a(0) \rangle - \Omega \langle a(0) \rangle$$

$$= -g \langle \hat{a}(\tau) a(0) \rangle - \gamma_{12} \langle \hat{a}^\dagger(\tau) \hat{a}(0) \rangle - \Omega \langle a(0) \rangle$$

$$\begin{aligned} \frac{d}{d\tau} \langle a(\tau) a(0) \rangle &= \sum_{j=k=1} M_{1j} \langle O_j(\tau) a(0) \rangle - \Omega \langle a(0) \rangle \\ &= -\gamma_{12} \langle a(\tau) a(0) \rangle - g \langle \hat{a}^\dagger(\tau) a(0) \rangle - \Omega \langle a(0) \rangle \end{aligned}$$

$$\langle a(\tau) a(0) \rangle = f(\tau)$$

$$\langle \hat{a}^\dagger(\tau) a(0) \rangle = g(\tau)$$

$$\begin{aligned} \dot{f}(\tau) &= -\gamma_{12} f(\tau) - g g(\tau) - \Omega a \\ \dot{g}(\tau) &= -\gamma_{12} g(\tau) - g f(\tau) - \Omega a \end{aligned}$$

$$\hookrightarrow \frac{1}{2} \left\{ \left(\alpha + \beta + \frac{2\Omega a}{\gamma_{12}} \right) e^{-(\gamma_{12}/2)\tau} + (\alpha - \beta) e^{(g \cdot \gamma_{12})\tau} - \frac{2\Omega a}{\gamma_{12}} \right\}$$

$$\frac{1}{2} \left\{ \left(\alpha + \beta + \frac{2\Omega a}{\gamma_{12}} \right) e^{-(\gamma_{12}/2)\tau} - (\alpha - \beta) e^{(g \cdot \gamma_{12})\tau} - \frac{2\Omega a}{\gamma_{12}} \right\}$$

(b) Set $g = \Omega = 0$ and consider the correlation function $G^{(2)}(\tau) = \langle a^\dagger(0) a^\dagger(\tau) a(\tau) a(0) \rangle$. What is the value of $G^{(2)}(0) = 2N_{th}^2$ for a thermal equilibrium state? Use the quantum regression theorem to derive an expression for $G^{(2)}(\tau)$ for arbitrary τ .

$$\rightarrow \hat{A} = \begin{pmatrix} \hat{a}^\dagger \\ \hat{a} \end{pmatrix} \rightarrow \dot{\hat{A}} = \begin{pmatrix} -\gamma_{12} & -g \\ -g & -\gamma_{12} \end{pmatrix} \hat{A} - \Omega$$

$$= \begin{pmatrix} -\gamma_{12} & 0 \\ 0 & -\gamma_{12} \end{pmatrix} \hat{A}$$

$$G^{(2)}(0) = \langle a^\dagger(0) a^\dagger(0) a(0) a(0) \rangle_{N_{th}} = \langle N_{th} | \hat{a}^\dagger(0) \hat{a}^\dagger(0) | N_{th} \rangle = 2N_{th}^2 = 2 \cdot \left(\frac{1}{e^{\frac{\hbar\omega}{kT}} - 1} \right)^2$$

$$G^{(2)}(\tau) = ?$$

16. Decay of a macroscopic superposition state (1P)

Consider a harmonic oscillator that is described by the master equation

$$\dot{\rho} = \frac{\gamma}{2}(N_{th} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{th}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger). \quad (1)$$

Assume that the harmonic oscillator is prepared in the pure state $\rho(t=0) = |\psi_0\rangle\langle\psi_0|$, where

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}(|n\rangle + |m\rangle) \quad (2)$$

is a superposition of two number states. Calculate the initial decay of the coherence $c_{n,m}(t) = \langle n|\rho(t)|m\rangle$, i.e., show that

$$\dot{c}_{n,m}(t) = -\frac{\Gamma_{n,m}}{2}c_{n,m}(t), \quad \text{for } t \rightarrow 0. \quad (3)$$

Derive the rate $\Gamma_{n,m}$.

$$\dot{\rho} = \frac{\gamma}{2}(N_{th} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{th}(2a^\dagger \rho a - a a^\dagger \rho - \rho a a^\dagger).$$

the harmonic oscillator is prepared in the pure state at $t=0$ is $|\psi_0\rangle\langle\psi_0|$ with

$$| \psi_0 \rangle \times \langle \psi_0 | = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \langle n | \hat{a}^\dagger = (\hat{a}^\dagger | n \rangle)^\dagger = (\sqrt{n+1} | n+1 \rangle)^\dagger = \sqrt{n+1} \langle n+1 |$$

$$\hat{a}^\dagger = \sqrt{n+1} | n+1 \rangle \langle n |$$

$$\hat{a}^\dagger = \sqrt{n+1} | n+1 \rangle \langle n |$$

$$\langle n | \hat{a}^\dagger = (\hat{a}^\dagger | n \rangle)^\dagger = (\sqrt{n+1} | n+1 \rangle)^\dagger = \sqrt{n+1} \langle n+1 |$$

$$\langle n | \dot{\rho}(t) | m \rangle = \frac{\gamma}{2}(N_{th} + 1) \left\{ 2 \langle n | a \rho \hat{a}^\dagger | m \rangle - \langle n | \hat{a}^\dagger \hat{a} \rho | m \rangle - \langle n | \rho \hat{a}^\dagger \hat{a} | m \rangle \right\} + \frac{\gamma}{2} N_{th} (\dots)$$

$$= 2\sqrt{n+1}\sqrt{m+1} \langle n+1 | \rho | m+1 \rangle - \sqrt{n} \langle n-1 | \hat{a} \rho | m \rangle - \sqrt{m} \langle n | \rho \hat{a}^\dagger | m-1 \rangle$$

$$= 2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - n \langle n | \rho | m \rangle - m \langle n | \rho | m \rangle$$

$$= 2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - (n+m) \rho_{n,m}$$

$$= \gamma_2(N_{th} + 1) \left(2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - (n+m) \rho_{n,m} \right) + \frac{\gamma}{2} N_{th} (2 \langle n | \hat{a}^\dagger \rho \hat{a} | m \rangle - \langle n | \hat{a} \hat{a}^\dagger \rho | m \rangle - \langle n | \rho \hat{a} \hat{a}^\dagger | m \rangle)$$

$$= 2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - \sqrt{n+1} \langle n+1 | \hat{a}^\dagger \rho | m \rangle - \sqrt{m+1} \langle n | \rho \hat{a} | m+1 \rangle$$

$$= 2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - (n+1) \langle n | \rho | m \rangle - (m+1) \langle n | \rho | m \rangle$$

$$= \gamma_2(N_{th} + 1) \left(2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - (n+m) \rho_{n,m} \right) + \frac{\gamma}{2} N_{th} (2\sqrt{n+1}\sqrt{m+1} \rho_{n+1,m+1} - (n+1) \rho_{n,m} - (m+1) \rho_{n,m})$$

$$= -\rho_{n,m} \left(\frac{\gamma}{2}(N_{th} + 1)(n+m) + \frac{\gamma}{2} N_{th} (n+1+m+1) \right) + \alpha \rho_{n+1,m+1} + \beta \rho_{n-1,m-1}$$

$$= -\rho_{n,m} \left(\frac{\gamma}{2} \left((N_{th} + 1)(n+m) + N_{th}(n+m+2) \right) \right) + \dots$$

$$= -\frac{\gamma}{2} \rho_{n,m} \left(N_{th}n + N_{th}m + n+m + N_{th}n + N_{th}m + 2N_{th} \right) + \dots$$

$$= -\frac{\gamma}{2} \rho_{n,m} \left((2N_{th} + 1)(n+m) + 2N_{th} \right) + \underbrace{\rho_{n+1,m+1} \alpha + \rho_{n-1,m-1} \beta}$$

$$\rightarrow \dot{c}_{n,m}(t) = -\frac{\Gamma}{2} c_{n,m} \quad \text{where}$$

$$\Gamma_{n,m} = \gamma \left((2N_{th} + 1)(n+m) + 2N_{th} \right)$$

$$| \psi_0 \rangle = \frac{1}{\sqrt{2}} (| n \rangle + | m \rangle)$$

$$\rightarrow \begin{aligned} \rho_{k,j} &\approx 0 \quad \text{for small } t \quad \text{for } k,j \neq n,m \\ \rho_{k,j} &= 0 \quad \text{for } t \rightarrow 0 \quad \text{for } k,j \neq n,m \end{aligned}$$